AccessLens v2:
Policy Access Risk Simulation and Explainable AI Interface

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Abstract

This progress report presents AccessLens v2, a system designed to model and simulate non-financial barriers in accessing government welfare schemes. Unlike predictive AI systems, AccessLens v2 employs a deterministic rule-based model combined with counterfactual simulations to evaluate policy interventions. The system operates on structured datasets containing scheme information, barrier taxonomies, and synthetic population profiles. AI is used exclusively for post-hoc explanation and language accessibility, ensuring transparency, reproducibility, and privacy preservation. This document describes the system architecture, dataset design, access risk model, counterfactual simulation engine, and preliminary results from batch simulations.

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1 Introduction

Government welfare schemes are designed to provide essential support to citizens, yet a significant gap persists between eligibility and actual access. While financial eligibility criteria receive considerable attention, non-financial barriers often determine whether entitled individuals successfully obtain benefits. These barriers include:

- **Awareness barriers**: Citizens may not know a scheme exists, understand its purpose, or be aware of application procedures.
- **Documentation barriers**: Required documents such as income certificates, residence proofs, or identity cards may be difficult or costly to obtain.
- **Digital access barriers**: Online-only application systems exclude those without internet connectivity, devices, or digital literacy.
- **Institutional dependency**: Reliance on intermediaries, local officials, or bureaucratic processes can create friction and opportunities for exclusion.

Understanding how these barriers interact and affect different population segments is essential for policy makers seeking to improve scheme reach. Simulation-based systems offer a valuable approach: they allow exploration of "what-if" scenarios without requiring real-world experimentation. By modeling access barriers and their effects, such systems can identify leverage points for intervention and support evidence-based policy discussions.

AccessLens v2 is designed to address this need. It provides a transparent, reproducible framework for simulating access barriers, evaluating policy interventions, and communicating results through explainable AI interfaces. This document reports on the current development progress and system architecture.

2 System Design Philosophy

AccessLens v2 is built on several foundational principles that distinguish it from conventional AI systems:

- **Deterministic modelling**: The core access risk model uses explicit rules and parameters rather than machine learning. This ensures that outputs are reproducible and behavior is fully explainable.
- **Transparency and interpretability**: All model components, parameters, and assumptions are documented and accessible. Users can inspect why a particular access risk score was computed.
- **Separation of simulation and explanation**: The simulation engine operates independently of AI. AI is used only after simulation to generate natural language explanations, never to determine outcomes.
- **Privacy preservation**: The system uses synthetic personas rather than real individual data. No personal information is collected, stored, or processed.
- **Counterfactual reasoning**: Policy analysis is conducted through systematic comparison of baseline scenarios with modified conditions, enabling exploration of intervention effectiveness.

These principles ensure that AccessLens v2 functions as a decision-support tool rather than an automated decision system, maintaining human oversight and accountability.

3 Dataset Design

AccessLens v2 operates on three structured datasets, each designed to capture different aspects of the access ecosystem.

3.1 Schemes Dataset (schemes_v0.csv)

The schemes dataset contains structured information about welfare schemes collected from official government sources and documentation. Each scheme is represented with the following attributes:

- `scheme_id`: Unique identifier for each scheme
- `scheme_name`: Official name of the scheme
- `domain`: Policy area (e.g., health, education, social security)
- `application_mode`: Primary application channel (online, offline, or hybrid)
- `documents_required_count`: Number of documents typically required
- `requires_income_certificate`: Binary indicator for income proof requirement
- `requires_residence_proof`: Binary indicator for residence proof requirement
- `digital_only`: Whether application is exclusively online
- `assisted_access_available`: Whether help desks or assisted channels exist
- `steps_count`: Approximate number of application steps
- `source_type`: Type of source document (gazette, scheme guidelines, etc.)
- `source_confidence`: Confidence level in data accuracy (high/medium/low)

Data collection prioritized publicly available information from official government portals, scheme guidelines, and published documentation.

3.2 Barrier Taxonomy Dataset (barriers_v0.csv)

The barrier taxonomy dataset represents different types of access barriers that can impede scheme uptake. Each barrier is categorized by type and assigned a default severity value representing baseline friction.

Barrier categories include:

- **Awareness**: Lack of information about scheme existence or procedures
- **Documentation**: Difficulty obtaining required paperwork
- **Digital**: Technology-related access barriers
- **Institutional**: Reliance on intermediaries or bureaucratic processes

Dataset fields:

- `barrier_id`: Unique identifier
- `barrier_type`: Category of barrier
- `affected_stage`: Stage in access process where barrier applies
- `default_severity`: Baseline friction value (0-1 scale)

3.3 Synthetic Personas Dataset (personas_v0.csv)

The synthetic personas dataset contains profiles used for simulation. These are constructed profiles representing different population segments, not real individuals. Using synthetic data ensures privacy while enabling meaningful analysis of how barriers affect different groups.

Persona attributes include:

- `persona_id`: Unique identifier
- `literacy_level`: Low, medium, or high (affects awareness barriers)
- `digital_access`: None, limited, or full (affects digital barriers)
- `document_completeness`: Fraction of required documents readily available
- `institutional_dependency`: Low, medium, high reliance on intermediaries

Synthetic personas are designed to represent realistic variation in the population while avoiding any association with actual individuals.

4 Access Risk Model

The core of AccessLens v2 is a deterministic model that computes access friction scores for each combination of scheme and persona. The model operates through three mechanisms:

- **Scheme barrier activation**: Each scheme specifies which barriers are relevant to its access process.
- **Persona amplification**: Persona attributes determine how strongly each barrier affects that individual.
- **Barrier severity values**: Each barrier has a base severity that can be modified by counterfactual conditions.

The conceptual formula for access risk is:

$$\text{AccessRisk} = \sum_{i=1}^{n} \left(\text{Severity}_i \times \text{Activation}_{i,\text{scheme}} \times \text{Amplification}_{i,\text{persona}}\right)$$

Where:

- n is the number of barrier types
- Severity_i is the baseline severity of barrier i
- $\text{Activation}_{i,\text{scheme}}$ is 1 if barrier i applies to the scheme, 0 otherwise
- $\text{Amplification}_{i,\text{persona}}$ captures how persona attributes affect susceptibility to barrier i

The raw score is normalized to a 0-1 range for interpretability, with higher scores indicating greater access difficulty.

5 Counterfactual Simulation Engine

The counterfactual simulation engine enables systematic evaluation of policy interventions by comparing baseline access risk scores with modified scenarios. Each counterfactual represents a specific policy change:

- **CF1: Awareness Improvement**: Reduces severity of awareness barriers, simulating the effect of information campaigns or outreach programs.
- **CF2: Documentation Simplification**: Reduces documentation barriers, representing policy changes that streamline paperwork requirements.
- **CF3: Assisted Access Removal**: Increases barriers for schemes that previously had assisted access, modeling what happens when support systems are withdrawn.
- **CF4: Digital Enablement**: Reduces digital barriers, simulating improved internet access, digital literacy programs, or device availability.

For each counterfactual, the system recalculates access risk scores with modified barrier severities. The difference between counterfactual and baseline scores (delta) indicates the potential impact of the intervention.

6 Batch Simulation and Analysis

The model was executed across all combinations of schemes and personas in the datasets, generating a comprehensive set of baseline risk scores. For each combination, the system then applied all four counterfactual scenarios and computed resulting deltas.

Simulation outputs were exported as structured CSV files containing:

- Scheme and persona identifiers
- Baseline access risk scores
- Counterfactual scores for each intervention
- Delta values (CF - baseline)

This batch approach enables systematic analysis of patterns across schemes, personas, and interventions. Results can be aggregated to identify which personas face highest barriers, which schemes present most difficulty, and which interventions show greatest promise.

7 Visualization and Results

Preliminary visualizations were generated from batch simulation outputs to identify patterns and insights:

- **Persona vs Scheme Risk Heatmap**: Reveals which persona types face highest barriers for which scheme categories. Shows concentration of access difficulty among vulnerable groups.
- **Average Risk by Persona**: Aggregates across schemes to show which persona attributes correlate with highest access friction. Indicates that digital access and document completeness are strong predictors.

- **Average Risk by Scheme**: Identifies scheme characteristics associated with high barriers. Digital-only schemes with multiple documentation requirements show consistently higher scores.
- **Counterfactual Effectiveness Ranking**: Compares average delta across interventions to identify which policy changes have largest potential impact.

Key insight from preliminary analysis: Removal of assisted access mechanisms (CF3) significantly increases access friction for personas with high institutional dependency. This suggests that support systems play a critical role in enabling access for vulnerable populations, and their removal could disproportionately harm those most in need.

8 AI Explanation Layer (Planned)

A planned extension of AccessLens v2 is the integration of AI for explanation and accessibility. This layer will operate exclusively on structured simulation outputs and will provide three capabilities:

- **Explainability**: AI will convert numerical simulation results into human-readable explanations. For example: "This persona faces high access difficulty primarily due to digital barriers, as the scheme requires online application and the persona has limited internet access."
- **Translation**: Explanations will be generated in multiple Indian languages including Hindi, Marathi, and Tamil, making system outputs accessible to diverse stakeholders.
- **Natural Language Interface**: Users will be able to query the system in plain language, such as "What happens if we simplify documentation for rural applicants?" The system will interpret the query, run appropriate simulations, and explain results.

Critically, the AI will only receive structured simulation outputs, not raw datasets. This maintains privacy while leveraging AI's communication capabilities.

9 Ethical Considerations

AccessLens v2 is designed with ethical considerations as foundational requirements:

- **No personal data**: The system uses only synthetic personas, eliminating privacy risks associated with real individual data.
- **No eligibility decisions**: The system models barriers and simulates outcomes but never determines who should receive benefits. All outputs are advisory.
- **Limited AI role**: AI is used only for explanation after simulation completes, never for decision-making or prediction.
- **Transparent assumptions**: All model parameters and assumptions are documented, enabling scrutiny and debate.
- **Advisory outputs**: System results are intended to inform policy discussions, not to automate or replace human judgment.

These safeguards ensure that AccessLens v2 contributes to responsible AI practice in governance contexts.

10 Future Work

Several extensions are planned for subsequent development phases:

- **Expanded scheme dataset**: Additional schemes across more domains and states will be added to increase coverage.
- **More detailed persona modelling**: Persona attributes will be refined and expanded to capture additional dimensions of vulnerability.
- **Improved counterfactual scenarios**: Additional intervention types and more nuanced policy parameters will be implemented.
- **Full AI interface integration**: The explanation layer will be fully implemented and tested with diverse user groups.
- **Interactive dashboard**: A web-based interface will enable stakeholders to explore simulations interactively.

These developments will transform AccessLens v2 from a research prototype into a practical tool for policy analysis and inclusive governance.